

Engineering Chemistry-1

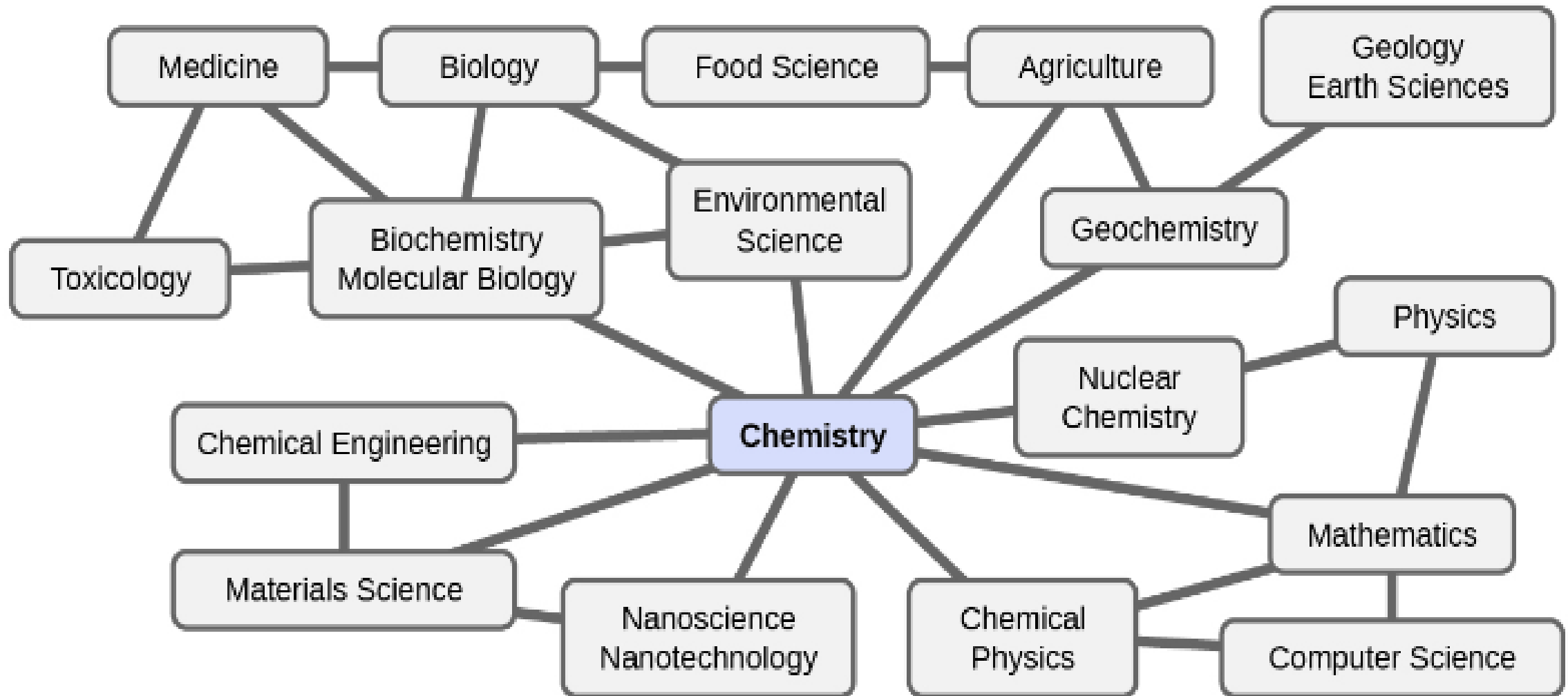
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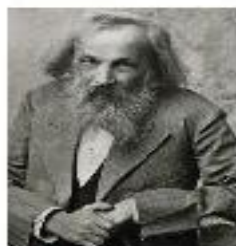
Dr. Zubair Hasan

Assistant Professor

Department of Mathematics and Physical Sciences (MPS)

Chemistry: The Central Science





Henri Becquerel

Ernest Rutherford

Niels Bohr

Heisenberg

Democritus

Mendeleev

-He proposed that all matter, including space and time, was made up of small units named atoms. He did no experiments and had little evidence, but his idea was kept on by Lucretius

-Created the periodic table of elements which organizes elements according to their similarities. His Period Law states that "physical and chemical of the elements are periodic functions of their atomic numbers."

-Discovered radioactivity, which earned him a Nobel Prize. Also discovered that rays emitted from uranium caused gases to ionize.

-Used the gold foil experiment to discover the modern model of an atom. He concluded that all positive charges were centralized, while negative electrons orbited the nucleus.

-Explained that outer orbits in an atom could hold more electrons than the inner orbits. By knowing this, one can determine the atom's chemical properties. He also gave birth to the idea that electrons emit light by jumping orbits.

-Thought of the Uncertainty Principle. This principle states that one can never know the exact location and energy of an electron simultaneously.

460BC

1803

1869

1885

1896

1897

1907

1910

1913

1925

1932

Dalton

-Agreed that all matter was created by atoms, which he believed were indestructible. He also stated that compounds are created by combining two atoms, and that all atoms of given elements are identical in their mass and properties.

Eugene Goldstein

-Discovered positive particles. He noted that the particles had a charge equal and opposite to the electron.

J.J. Thomson

-Discovered the electron. He experimented by testing and studying the nature of electric discharge in a high vacuum cathode-ray tube.

Millikan

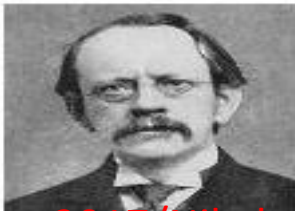
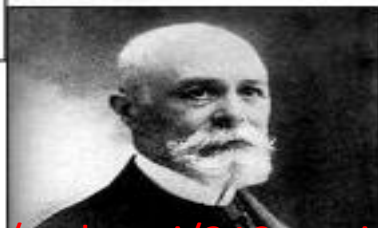
-Used an oil drop experiment to determine the charge of a single electron.

Frederick Soddy

-Came up with the concept of isotopes. Explain with Ernest Rutherford that radiation is due to transmutation of elements.

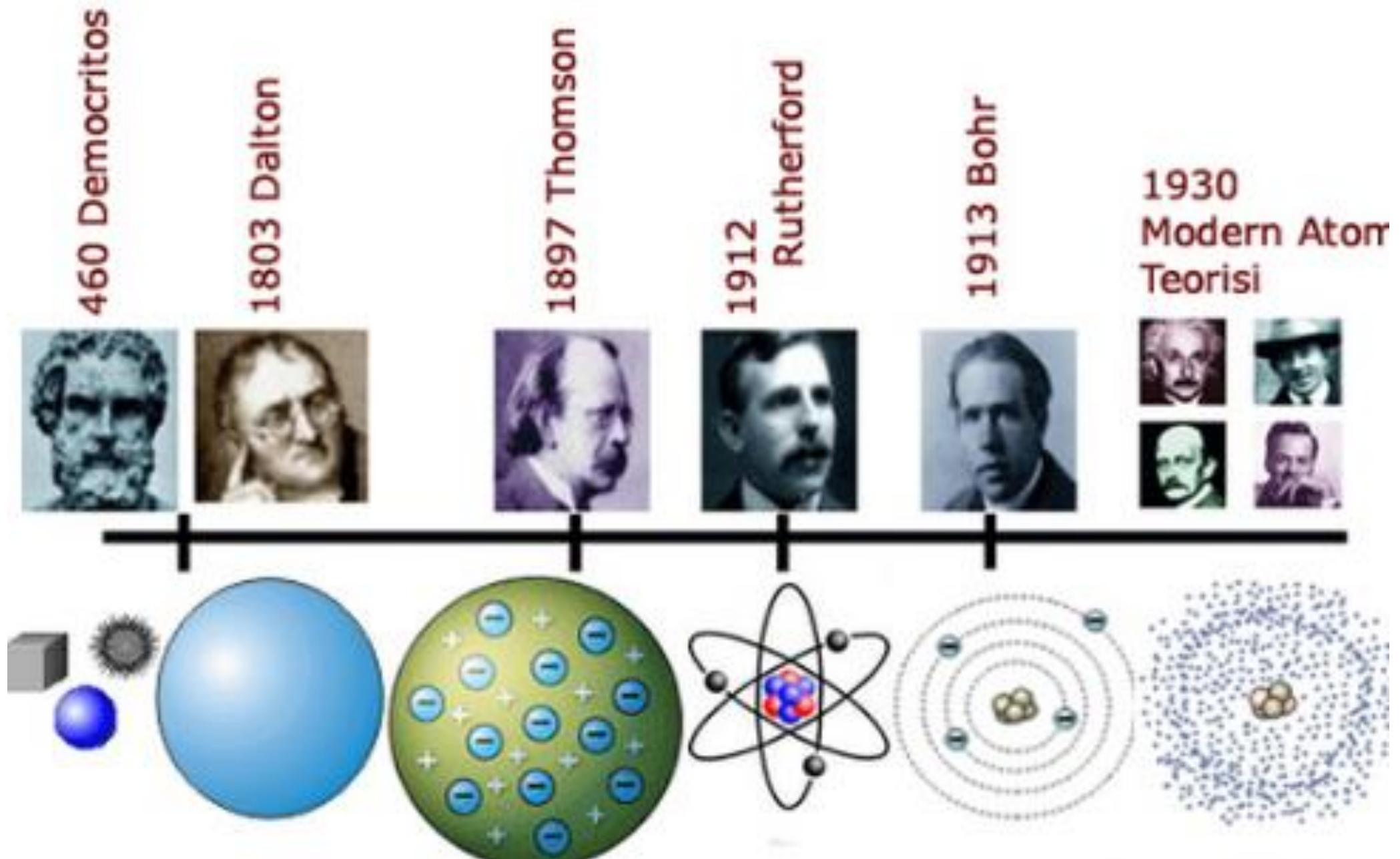
Chadwick

-Discovered the neutron. The neutron helps balance out protons in the nucleus of an atom.



History of the Atom

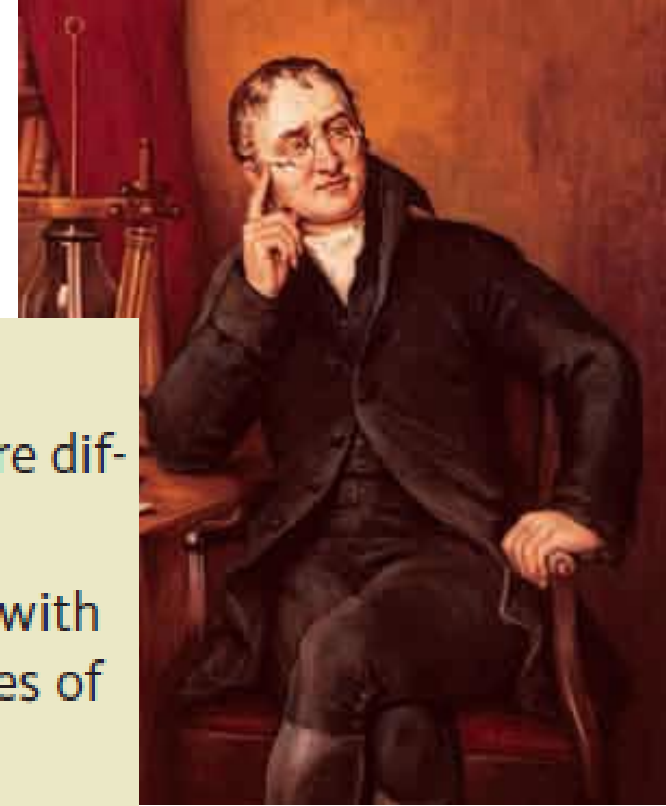
By Matt Rosa



Atomic Structure

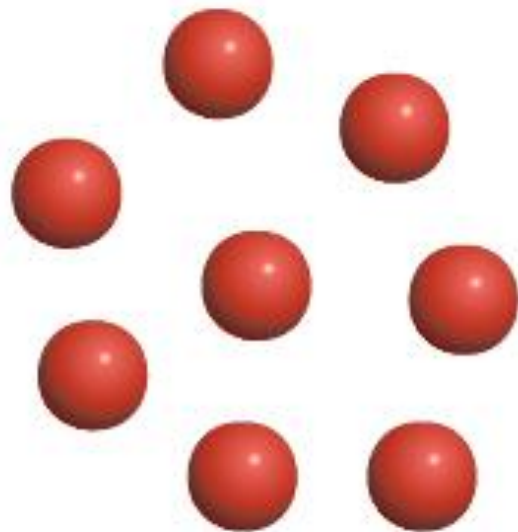
In 1808 Dalton published A New System of Chemical Philosophy, in which he presented his theory of atoms:

1. Each element is made up of tiny particles called atoms.
2. The atoms of a given element are identical; the atoms of different elements are different in some fundamental way or ways.
3. Chemical compounds are formed when atoms of different elements combine with each other. A given compound always has the same relative numbers and types of atoms.
4. Chemical reactions involve reorganization of the atoms—changes in the way they are bound together. The atoms themselves are not changed in a chemical reaction.

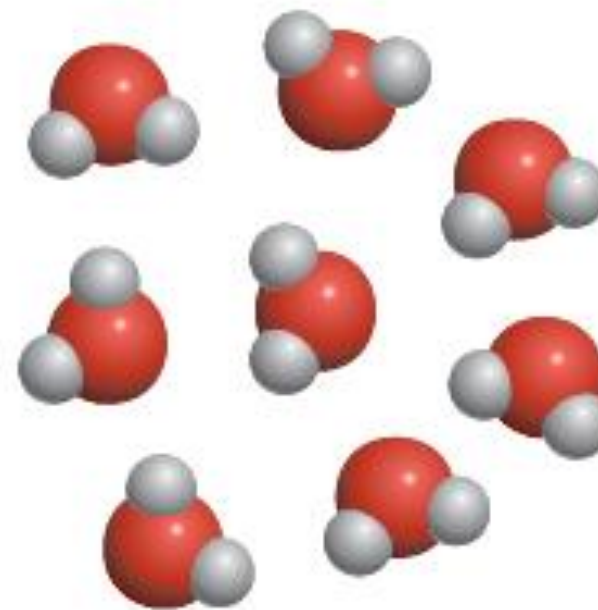




Atoms of element X



Atoms of element Y



Compounds of elements X and Y

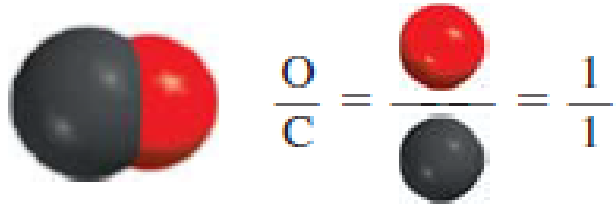
According to Dalton's atomic theory, atoms of the same element are identical, but atoms of one element are different from atoms of other elements.

Compound formed from atoms of elements X and Y. In this case, the ratio of the atoms of element X to the atoms of element Y is 2:1. Note that a chemical reaction results only in the rearrangement of atoms, not in their destruction or creation.

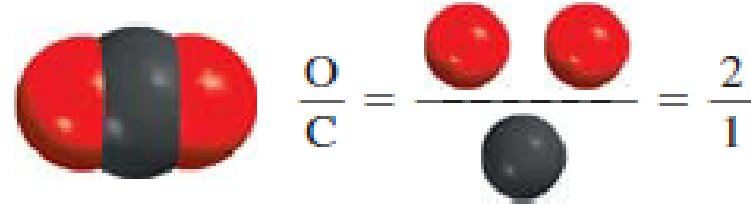
The third hypothesis is an extension of Proust's law (1799) of **definite proportions**:

“Different samples of the same compound always contain its constituent elements in the same proportion by mass”

Carbon monoxide



Carbon dioxide



The third hypothesis also supports the **law of multiple proportions**:

“if two elements can combine to form more than one compound, the masses of one element that combine with a fixed mass of the other element are in ratios of small whole numbers”

Practice Problem

- Two different compounds are formed by the elements carbon and oxygen. The first compound contains 42.9% by mass carbon and 57.1% by mass oxygen. The second compound contains 27.3% by mass carbon and 72.7% by mass oxygen. Show that the data are consistent with the law of multiple proportions.
- Show that Nitrogen monoxide (NO) and nitrogen dioxide (NO_2) satisfy the law of multiple proportions.
- Hydrazine, ammonia, and hydrogen azide all contain only nitrogen and hydrogen. The mass of hydrogen that combines with 1.00 g of nitrogen for each compound is 1.44×10^{-1} g, 2.16×10^{-1} g, and 2.40×10^{-2} g, respectively. Show how these data illustrate the law of multiple proportions.

Dalton imagined an atom that was both extremely small and indivisible.
How long this imagination exists

A series of investigations that began in the 1850s and extended into the twentieth century clearly demonstrated that atoms actually possess internal structure; that is, they are made up of even smaller particles, which are called **subatomic particles**. This research led to the discovery of three such particles

Electrons, protons, and neutrons.

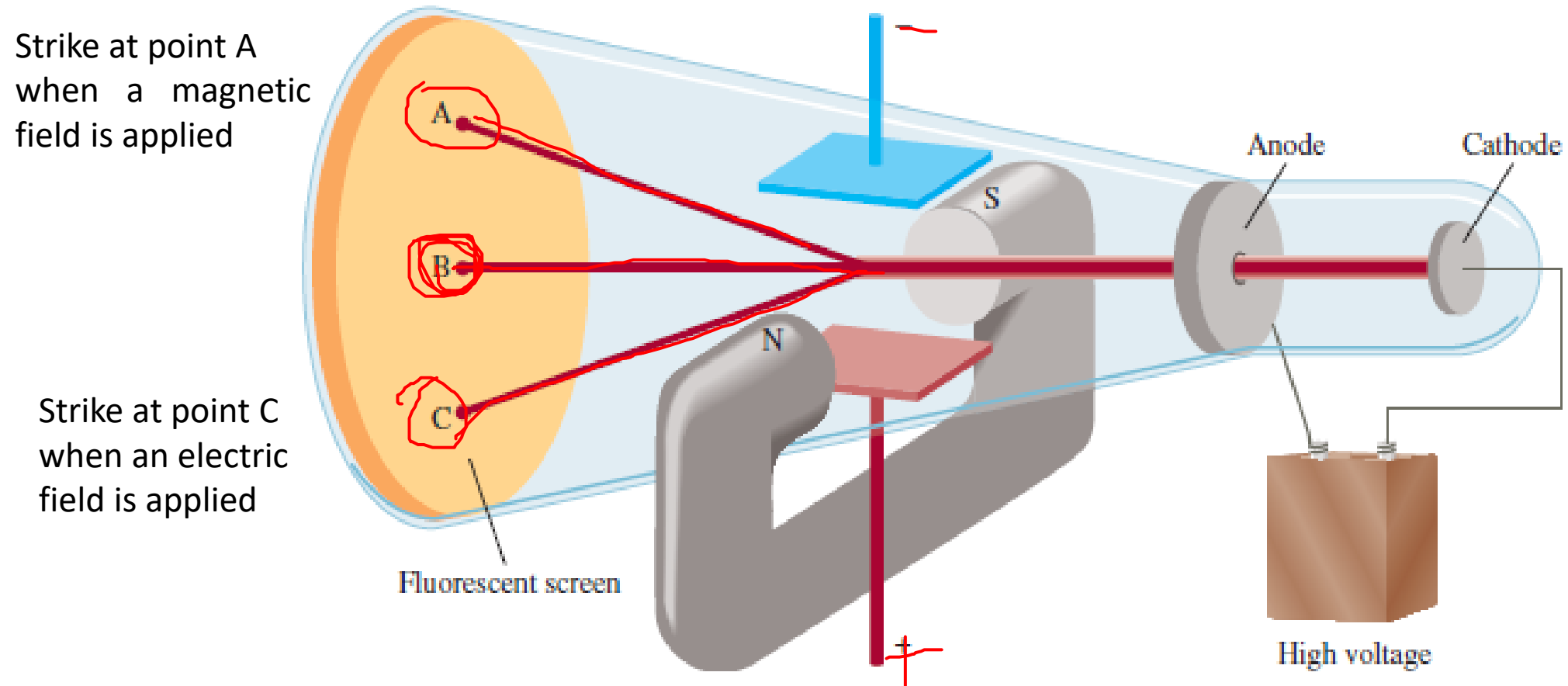
The Electron

The first important experiments that led to an understanding of the composition of the atom were done by the English physicist J. J. Thomson.

Research area:
electrical discharges in partially evacuated tubes called cathode-ray tubes.



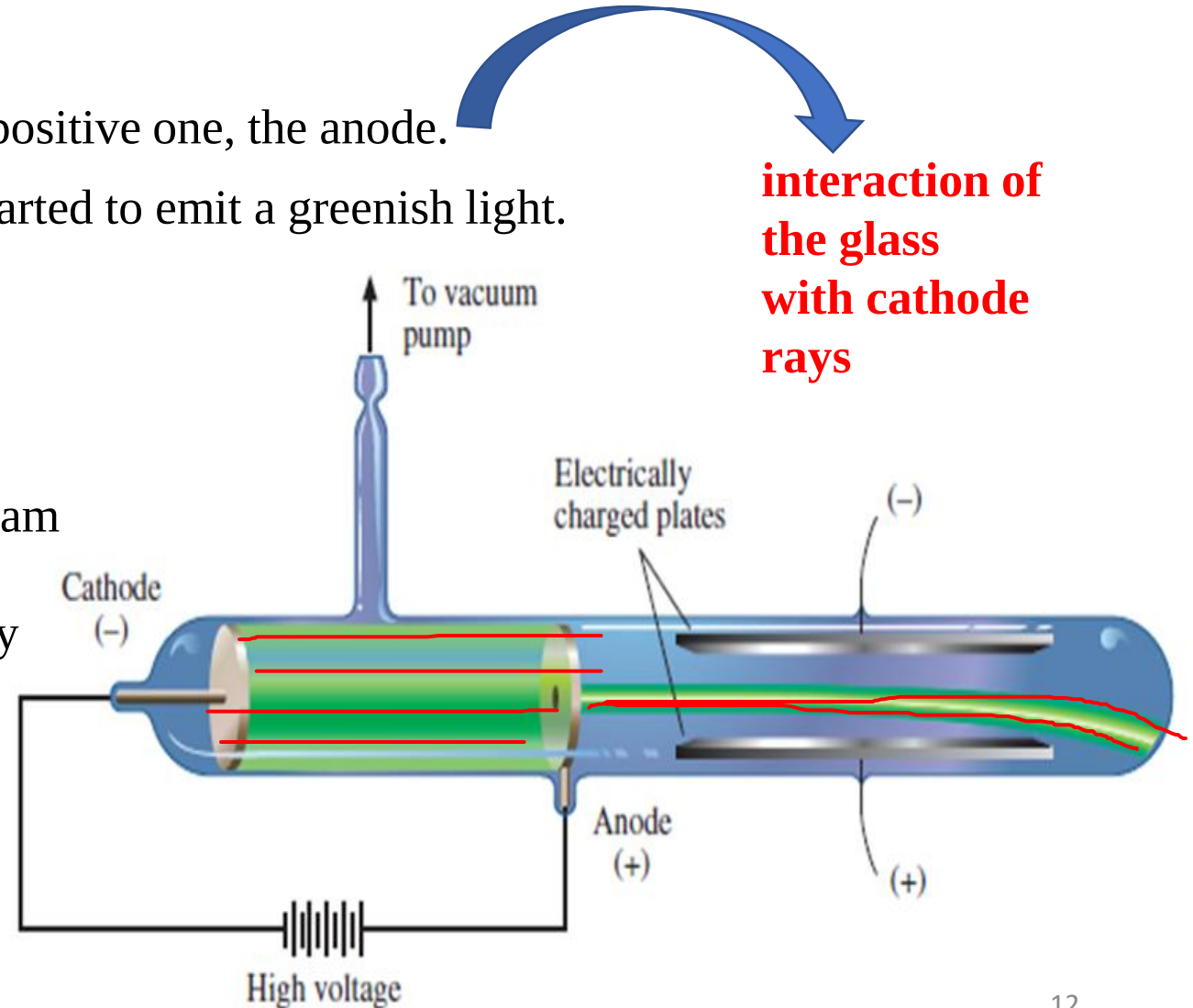
Joseph John Thomson
(1856–1940)



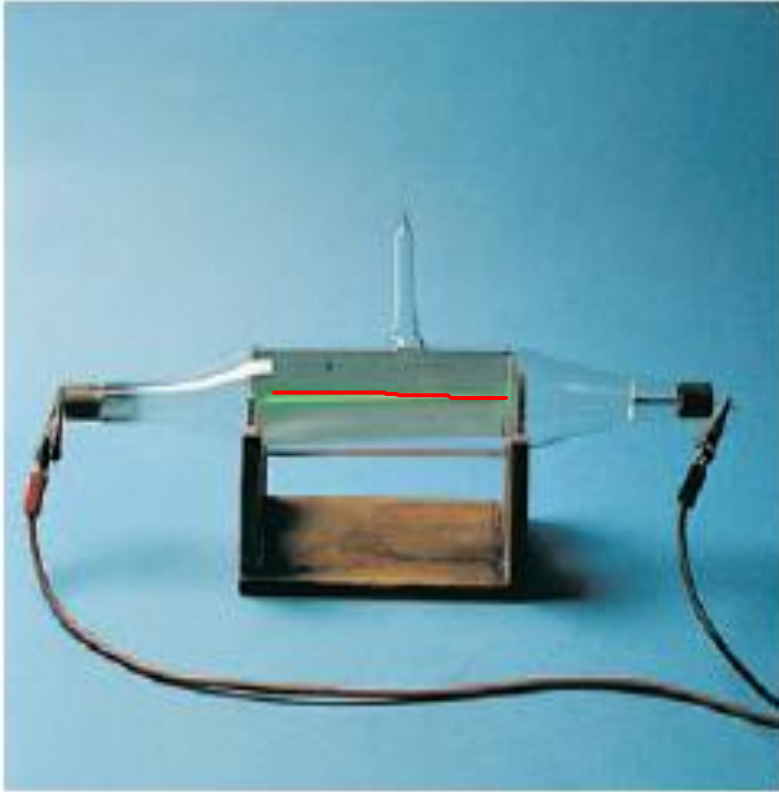
A typical cathode ray tube with an electric field perpendicular to the direction of the cathode rays and an external magnetic field.

Discovery of the Electron

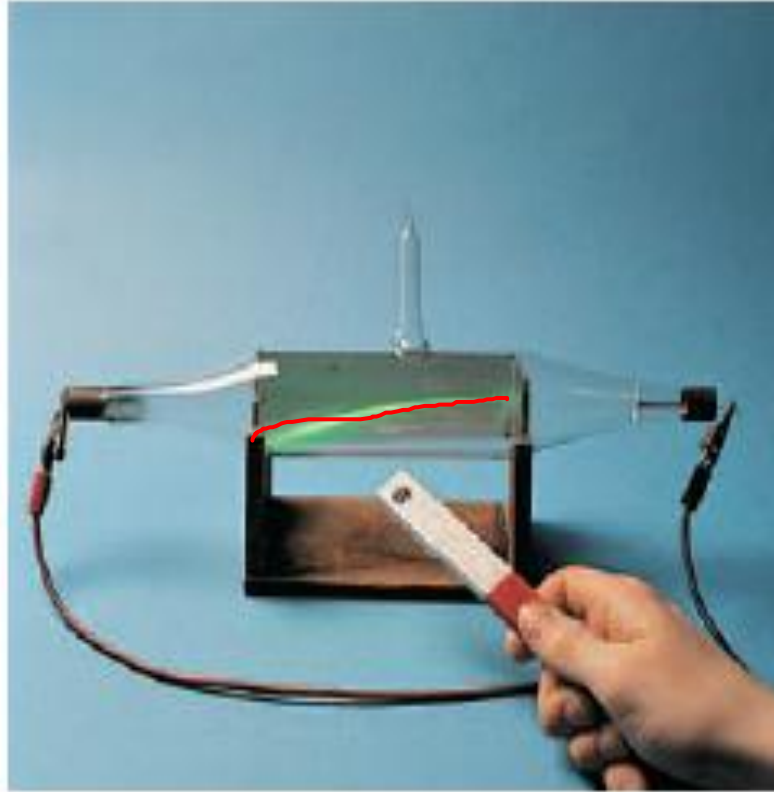
- In this apparatus, two electrodes from a high-voltage source are sealed into a glass tube from which the air has been evacuated.
- The negative electrode is called the cathode; the positive one, the anode.
- Turning on high-voltage current, the glass tube started to emit a greenish light.
- Cathode rays travel from the negative cathode to anode.
- When some rays pass through a hole to form a beam (Figure 2.4). This beam bends away from the negatively charged plate and toward the positively charged plate.



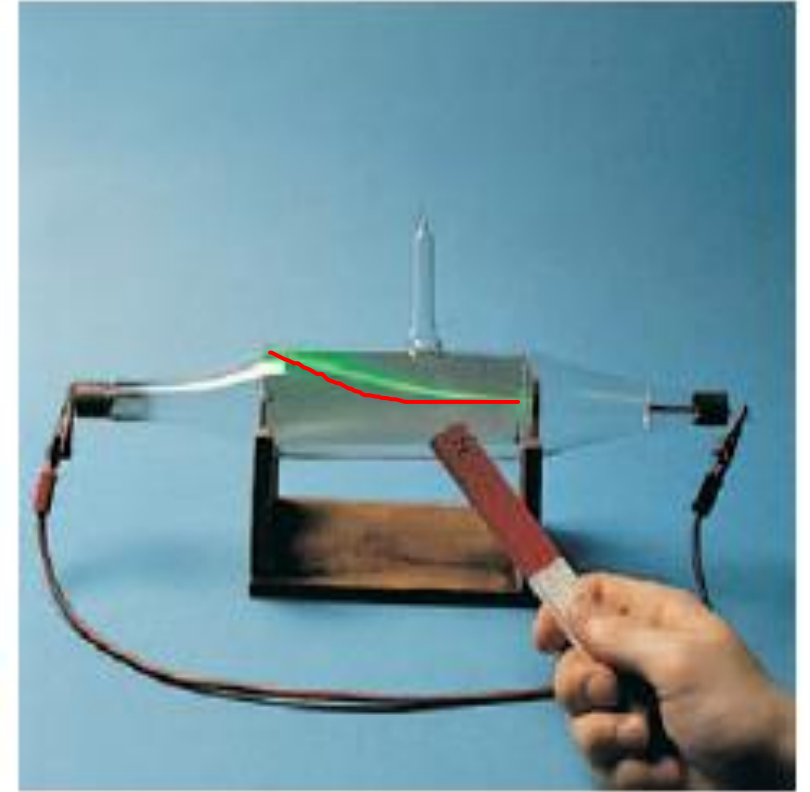
Bending of cathode rays by a magnet



The cathode-ray beam travels from right to left. It is visible where it falls on a zinc-sulfide screen.



The beam of negative particles bends downward as the south pole of the magnet is brought toward it.



When the magnet is turned around, the beam bends in the opposite direction.

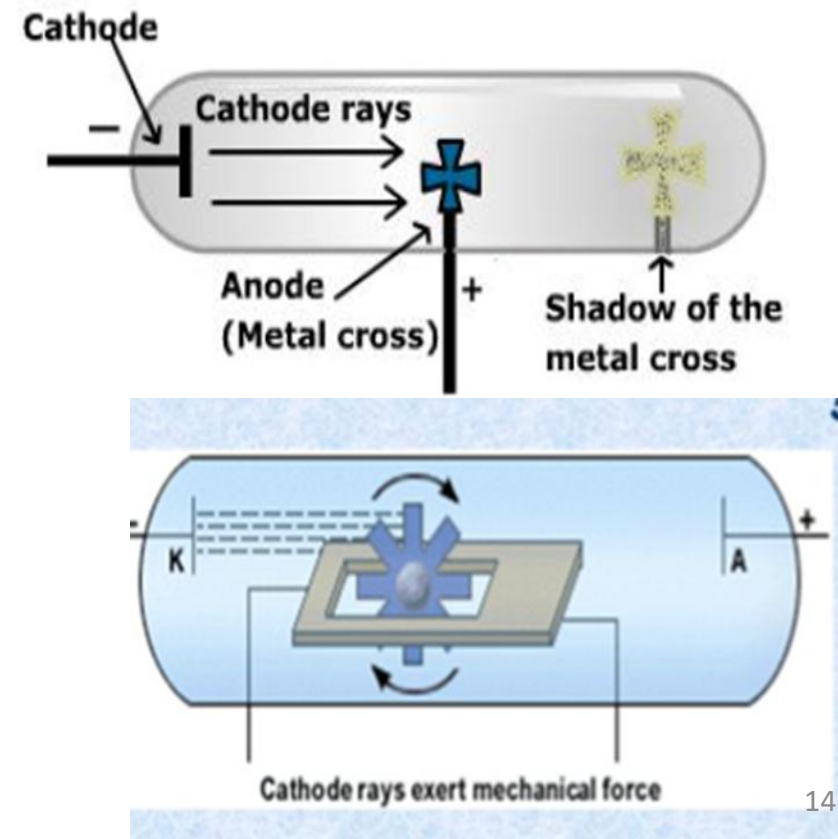
Findings:

Thomson showed that the characteristics of cathode rays are independent of the material making up the cathode.

From such evidence, he concluded that a cathode ray consists of a beam of negatively charged particles (or electrons) and that electrons are constituents of all matter.

Properties of Cathode Ray:

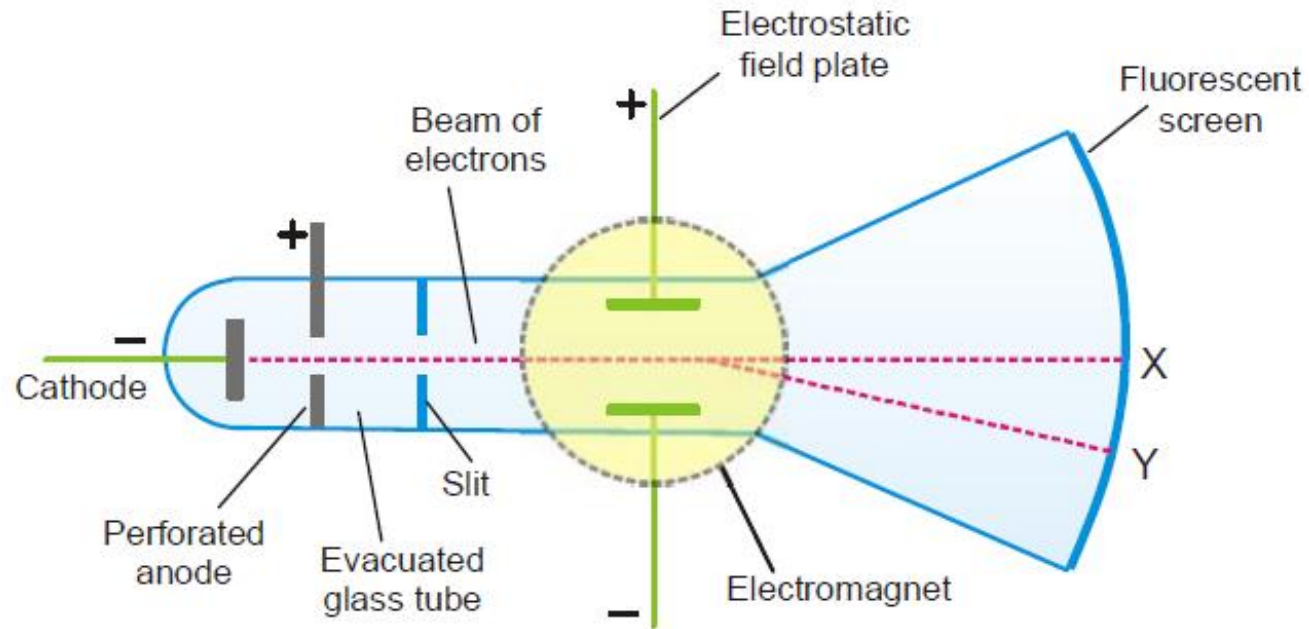
- They travel in straight lines away from the cathode and cast shadows of metallic objects placed in their path.
- Cathode rays cause mechanical motion of a small pin-wheel placed in their path. Thus they possess kinetic energy and must be material particles.



Properties of Cathode Ray:

- They produce fluorescence (a glow) when they strike the glass wall of the discharge tube.
- They heat up a metal foil to incandescence which they impinge upon.
- Cathode rays produce X-rays when they strike a metallic target.
- Cathode rays are deflected by the electric as well as the magnetic field in a way indicating that they are streams of minute particles carrying negative charge.

Measurement of e/m for electrons



Magnetic field is applied first and causes the electrons to be deflected in a circular path while the spot is shifted to Y.

The radius of the circular path can be obtained from the dimensions of the apparatus, the current and number of turns in the coil of the electromagnet and the angle of deflection of the spot.

An electrostatic field of known strength is then applied so as to bring back the spot to its original position.

Then from the strength of the electrostatic field and magnetic field, it is possible to calculate the velocity of the electrons.

B = magnetic field strength; v = velocity of electrons
 e = charge on the electron; m = mass of the electron
 r = radius of the circular path of the electron in the magnetic field.

$$Bev = \frac{mv^2}{r}$$

r is obtained from the dimensions of the tube and the displacement of the electron spot on the fluorescent screen.

When the electrostatic field strength and magnetic field strength are counterbalanced

Where E is the strength of the electrostatic field.

If E and B are known, v can be calculated and on substitution in equation (1), we get the value of e/m .

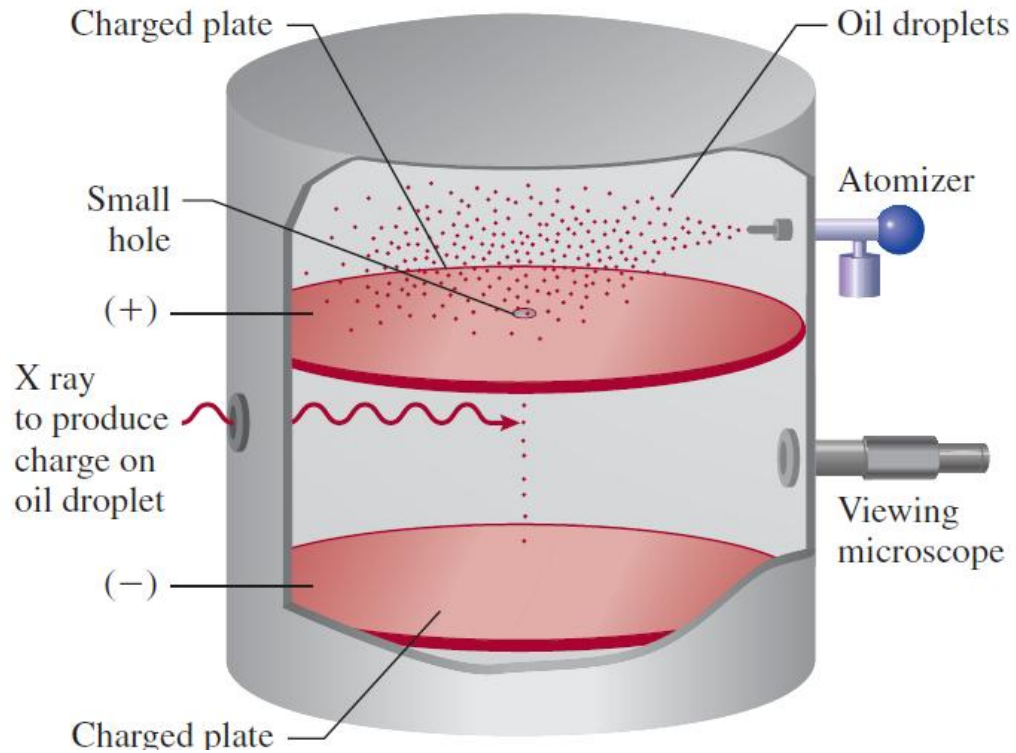
$$Bev = Ee$$

$$v = \frac{E}{B}$$

$$\frac{e}{m} = \frac{E}{B^2 r}$$

e/m for the electron = -1.76×10^8 coulomb/g

Millikan oil drop experiment



His work proved that the charge on each electron was exactly the same. In his experiment, Millikan examined the motion of single tiny drops of oil that picked up static charge from ions in the air. He suspended the charged drops in air by applying an electric field and followed their motions through a microscope. Using his knowledge of electrostatics, Millikan found the charge of an electron to be $1.6022 \times 10^{-19} \text{ C}$. From these data he calculated the mass of an electron:

$$\begin{aligned} \text{mass of an electron} &= \frac{\text{charge}}{\text{charge/mass}} \\ &= \frac{-1.6022 \times 10^{-19} \text{ C}}{-1.76 \times 10^8 \text{ C/g}} \\ &= 9.10 \times 10^{-28} \text{ g} \end{aligned}$$

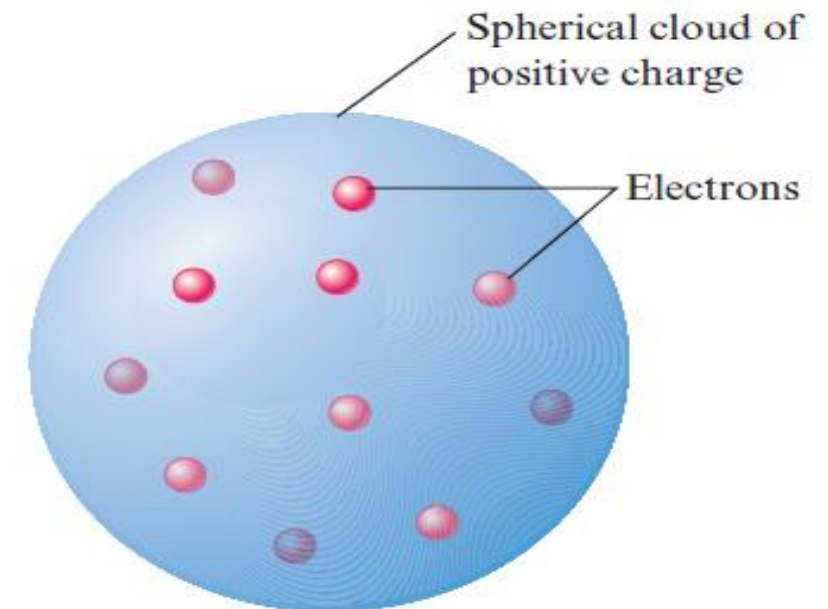
Atomic Model: Plum pudding model

- One of goals of cathode-ray tube experiments was to gain an understanding of the structure of the atom.
- He reasoned that since electrons could be produced from electrodes made of various types of metals, all atoms must contain electrons.
- Since atoms were known to be electrically neutral, Thomson further assumed that atoms also must contain some positive charge.
- Thomson postulated that an atom consisted of a diffuse cloud of positive charge with the negative electrons embedded randomly in it.
- This model, shown in Fig. is often called the plum pudding model because the electrons are like raisins dispersed in a pudding (the positive charge cloud), as in plum pudding, a favorite English dessert.

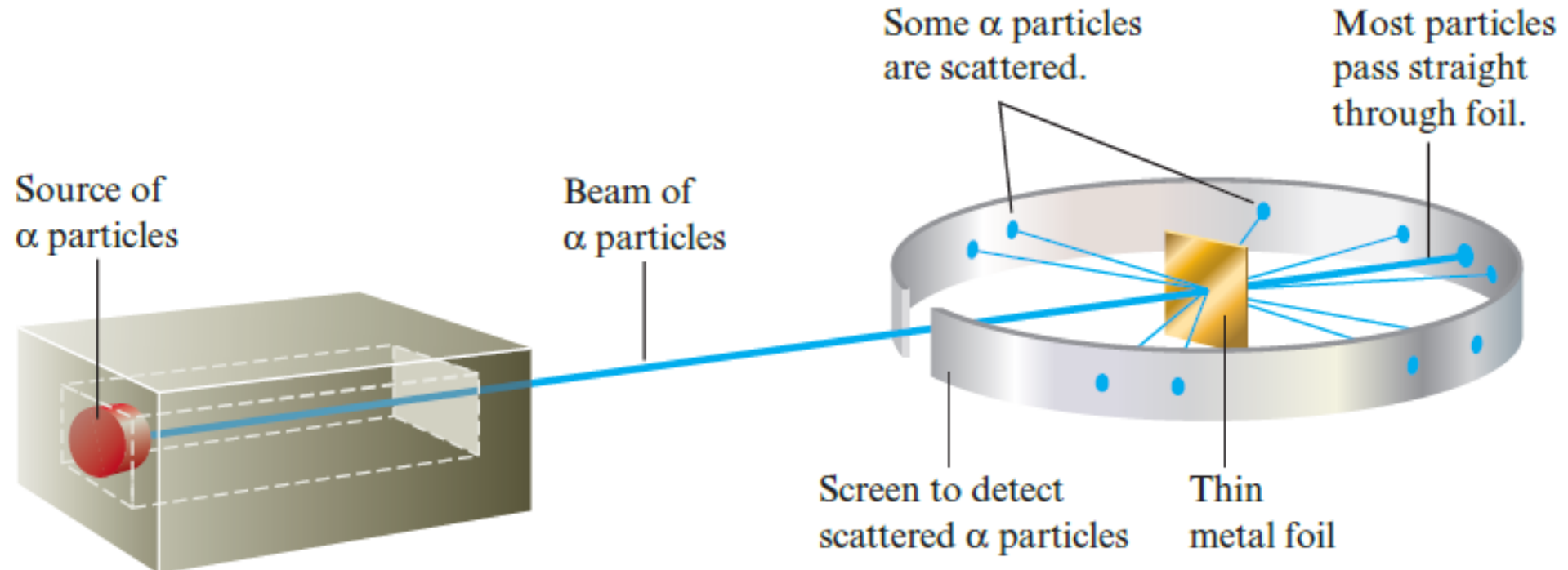


StockFood/Getty Images

A classic English plum pudding in which the raisins represent the distribution of electrons in the atom.



Alpha Particle-Scattering Experiment



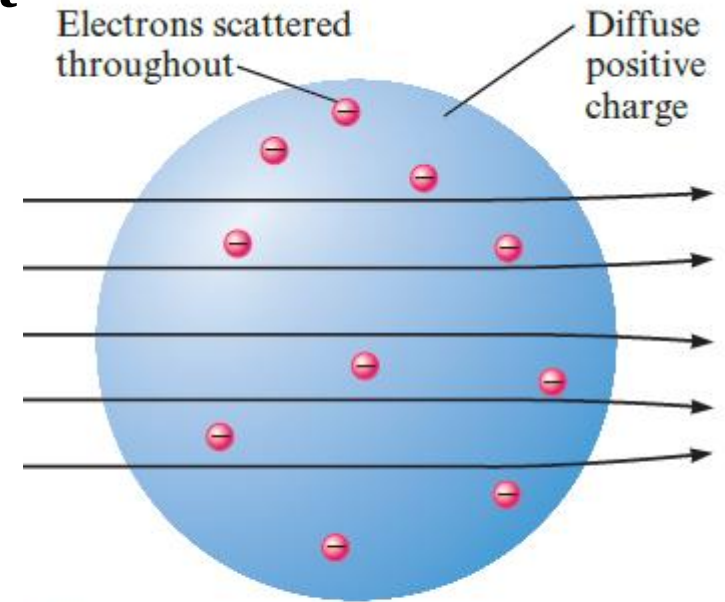
They directed a stream of very highly energetic α -particles from a radioactive source against a thin gold foil provided with a circular fluorescent zinc sulphide screen around it. Whenever an α -particle struck the screen, a tiny flash of light was produced at that point

Alpha Particle-Scattering Experiment

If Thomson's model were accurate, the massive α particles should crash through the thin foil like cannonballs through Gauze.

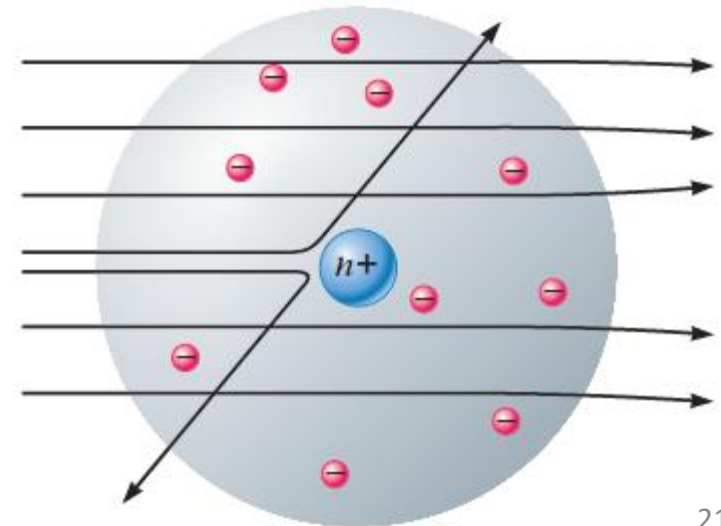
He expected the α particles to travel through the foil with, at the most, very minor deflections in their paths.

The results of the experiment were very different from those Rutherford anticipated. Although most of the α particles passed straight through, many of the particles were deflected at large angles, as shown in Fig. (b), and some were reflected, never hitting the detector.



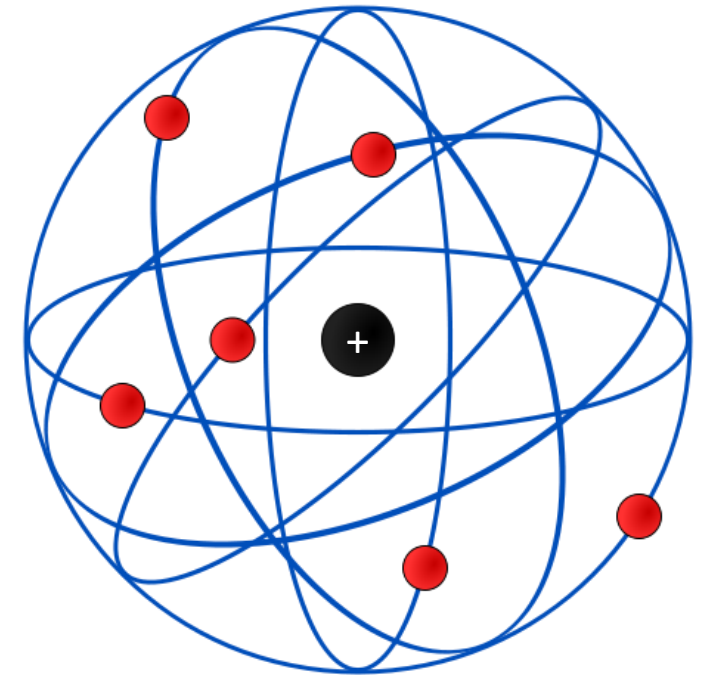
a

The expected results of the metal foil experiment if Thomson's model were correct.



Postulates of Rutherford atomic model:

- Atom has a tiny dense central core or the nucleus which contains practically the entire mass of the atom, leaving the rest of the atom almost empty.
- The entire positive charge of the atom is located on the nucleus, while electrons are distributed in vacant space around it.
- The electrons were moving in orbits or closed circular paths around the nucleus like planets around the sun.

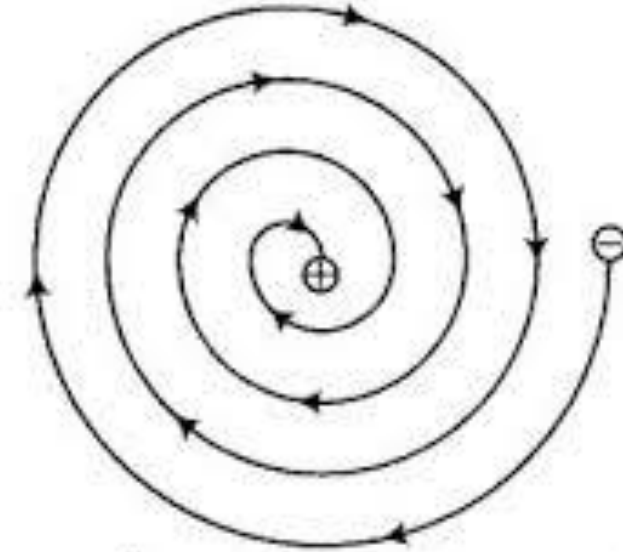


Weakness of Rutherford Atomic Model

According to the classical electromagnetic theory if a charged particle accelerates around an oppositely charged particle, the former will radiate energy.

If an electron radiates energy, its speed will decrease and it will go into spiral motion, finally falling into the nucleus.

This does not happen actually as then the atom would be unstable which it is not. This was the chief weakness of Rutherford's Atomic Model.



Continuous loss of energy by a revolving electron